TRIAL AT PORTO NOVO IRON WORKS

The next trial of which I have any notes was made at Porto Novo. Here, a pair of arms was fitted to a common boiler, the evaporating power of which was not measured, but was estimated by the engineer (Mr Brunton) to be sufficient to supply a 5 horsepower engine in the common way, that is, evaporating about 7 or 8 cubic feet of water per hour. At an effective pressure of 50 lbs, the arms, which were each 30 inches long, and of a circular form 1¼ inches diameter, attained a speed of 1,000 revolutions per minute, indicating a velocity at their extremities of 15,708 feet, without any load. [15]

> The next memorandum which I come to is a valuable experiment on the issue of steam. It is contained in the following extract dated 5 May 1838.

Experiment on 5 May 1838:

> By a very careful experiment continued for an hour and thirty minutes, I obtained a constant pressure at the end of the arms of 2¼ lbs at 40 lbs (effective) pressure per square inch—diameter of orifices $^{3}/_{16}$ inch each, or area of both $^{1}/_{12}$ square inch.

The experiment was repeated the same month with another pair of arms and it worked very well.

Experiment on 31 May 1838:

> On 31 May another pair of arms was attached to a large boiler, whose evaporating power was estimated to be equal to about 41 cubic feet per hour; there was therefore calculated to remain 10 cubic feet per hour for the supply of the arms. The greatest velocity obtained by this engine was from 15 to 18,000 feet per minute; and by way of trial, an iron lathe was attached to it, which ‘worked very well’.

Finally, a locomotive!

> While these experiments were being made, another small engine was in progress at the Red Hills. It was originally constructed as a locomotive, to run on the railway between that point and Madras, but its power was found insufficient for that purpose.
>
> Experiments were conducted on this engine and sufficient knowledge was derived from the results obtained to show that the evaporating power of a boiler capable of keeping up the required force at the end of the arms must be much greater than the one tried; and it also seemed to be a fair conclusion from the whole of the trials made, that 8–10 oz. per cubic foot of water evaporated per hour, was the utmost that could be safely calculated on.
>
> A new locomotive engine was now constructed from the above data; and, as it was considered that a force of 3 lbs at the end of the arms would be amply sufficient for the purpose required, such a capacity was given to the boiler as to ensure an evaporating power